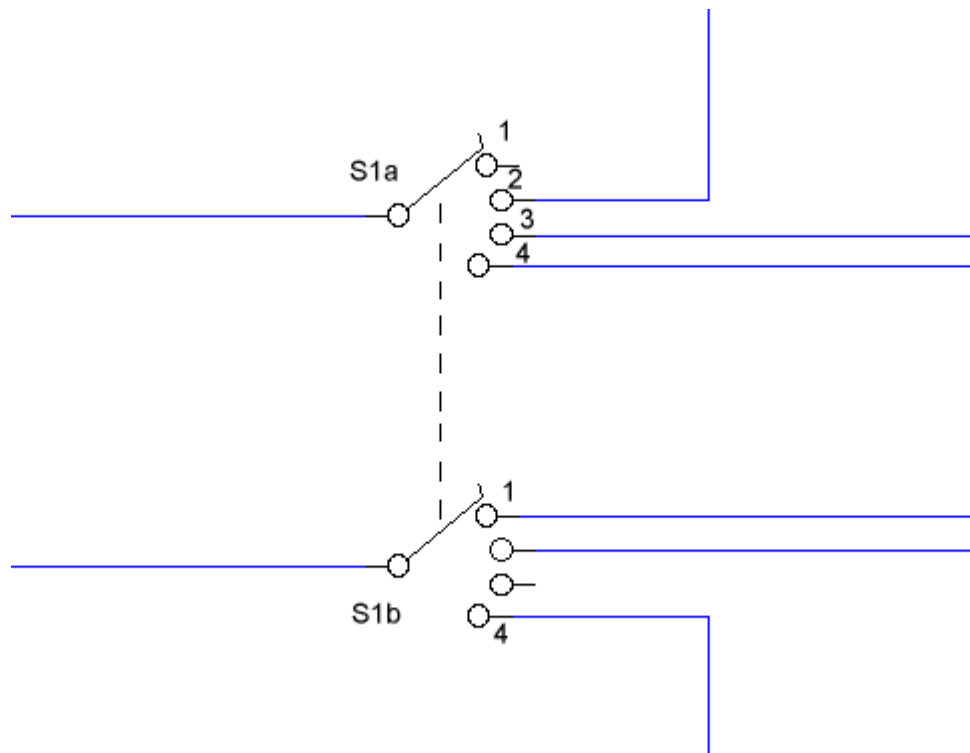


Switch Wiring and Relay Contacts

For anyone who has ever had problems wiring relays or switch contacts then this guide may help.

Switches

Switches are pretty straight forward to wire, but problems may arise with multi-pole multi-way switches. Switch contacts are always drawn in the OFF position. In the case of rotary switches the wiper arm will always be connected to the first contact which will be the off position. The diagram below shows a 2 pole 4 way switch.

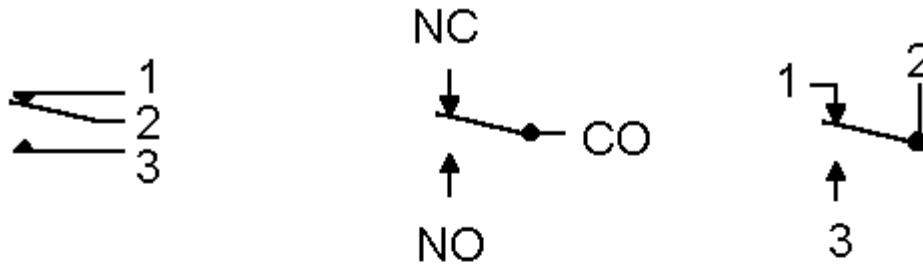


The first set of contacts or *arc* is designated S1a. The contacts are numbered 1 through 4, as switches are always drawn in the off position then the contact connected to the wiper arm will be the OFF position. In the diagram the contacts are labeled 1 through 4 for clarity though with no labeling it is easy to identify the first contact. Arc b is shown below. It is common practice in schematic drawings to link both switch arcs with a *dashed line* as shown above. The contact labeled 1 is the switches first position and contact 3 is not used. With all multi pole switches the wiper arm will track, so that if the switch is turned from off two positions then both wiper arms on S1a and S1b would rest on contact 3.

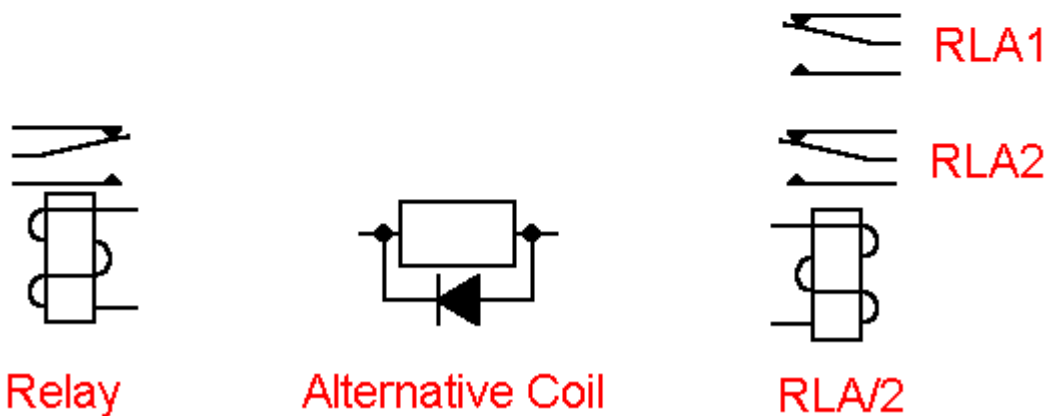
Relays and Contacts

As with switches, relays are always drawn in the off or *unoperated position*. Contacts may be drawn in different ways as shown below; but the middle contact (labeled 2 below) is the wiper arm and will move when the relay is energized. This contact is called the *changeover contact*, abbreviated CO. The top contact

(labeled 1) is the normally closed, NC contact (relay not energized) and contact 3 known as normally open, NO. When power is applied to the relay coil, the NO contact is then mechanically and electrically contacted to the CO contact.



Contacts



Relay

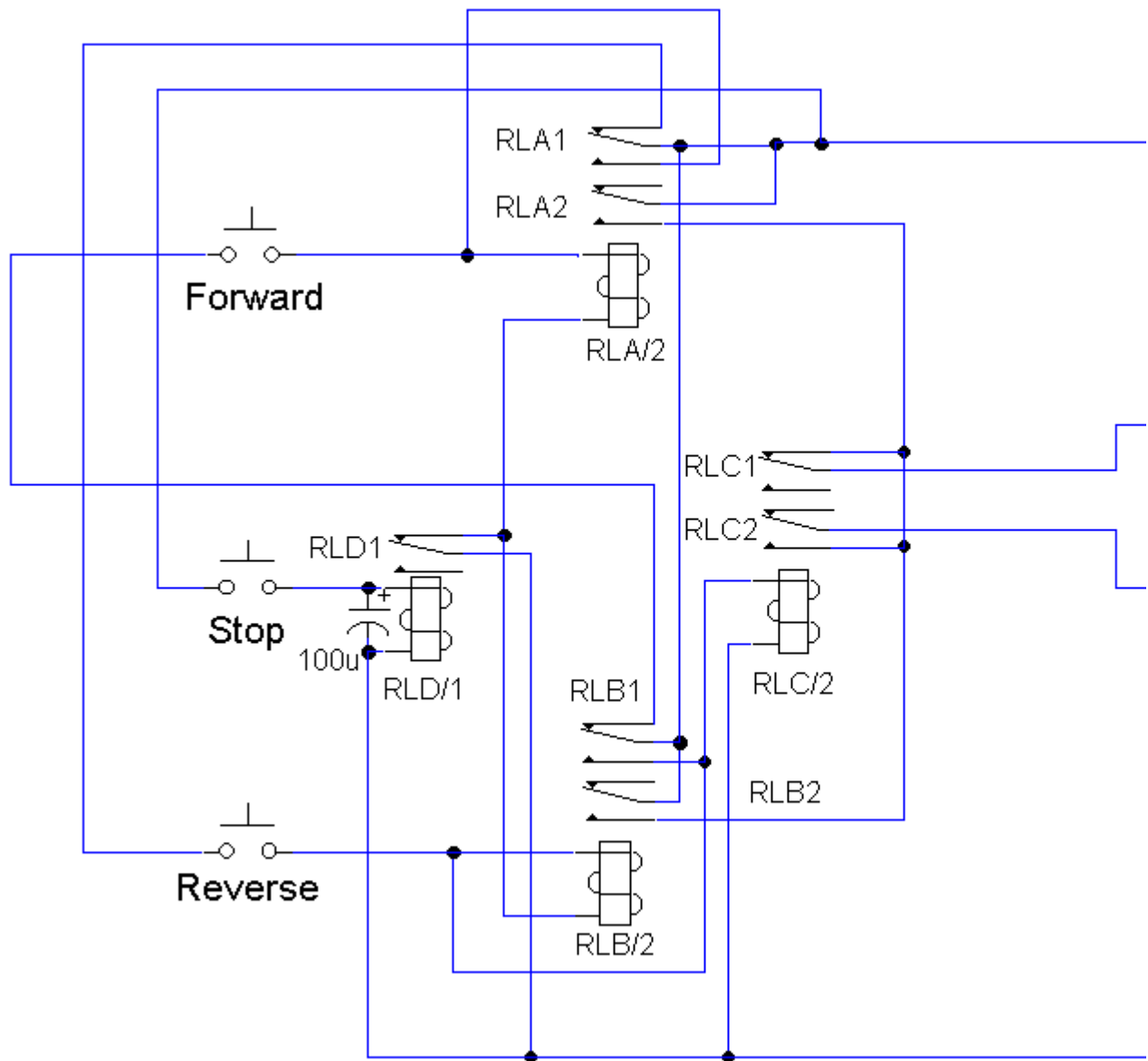
Alternative Coil

RLA/2

Relay Contact Labeling

As shown above, bottom right hand corner the relay has two changeover contacts. The designation for a relay coil may be, RL or just R. If there is more than one relay in a circuit then the usual method is to designate relays as RLA, RLB, or RA, RB etc. The easiest way to keep track of how many contacts belong to a particular relay coil is to use a denominator with the amount of contacts. In this case RLA/2 has two contacts and these are appropriately labeled RLA1 and RLA2. Easy. Now for a motor reversing circuit using non latching press button switches.

Motor Reversing Circuit



This circuit uses 4 relays and 3 push buttons for Stop, Forward and Reverse control of a DC motor. RLD has just one changeover contact whereas RLA, RLB and RLC have 2 CO contacts each. Basically RLD will stop the motor, RLA latches the motor in a forward direction and RLB latches the motor in the opposite direction, whilst RLC and its contacts provide the reversing for the motor. This circuit is quite messy and it is common practice to see the relay coil and its associated contacts separated some distance in the actual schematic. The [following circuit](#) is identical (electrically) and you should now be able to see that using the above rules.

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